CS532 Quiz 5: Structure from Motion

Name: _____

Question 1. You're given the following two real-world capture setups. For each scenario, state (and justify in 2–3 sentences) whether the additional geometric information is sufficient to upgrade a projective reconstruction on the entire scene. If it is possible to upgrade, state upto which reconstruction you can upgrade to for each of these scenarios. If you think this information is not enough to get you metric reconstruction for the 3D scene (upto scale), what extra information is required?

1. **Scenario A (Office facade):** You walk past a building facade and capture it from a handheld camera. The building facade has clearly parallel horizontal bands. You also see parallel pedestrian street crossing lines on the road.

   Solution: We cannot upgrade this to affine since we only get 2 points at infinity from the 2 sets of parallel lines lying on 2 different planes. We need 3 points to estimate plane at infinity and ths upgrade to affine. Knowing another set of vertical lines (eg. building edges) will give us this constraint.

2. **Scenario B (tiled kitchen floor):** You're filming in your home kitchen. The floor is planar and covered with square tiles laid out in a regular grid.

   Solution: Here we can get a similarity upgrade since we know the 2 sets of parallel lines and also the angles between them. So we can calibrate the camera and get a similarity reconstruction. If we know the size of the tiles, we can get the metric reconstruction. (Note that this assumes the floor has good coverage from different viewpoints).

Question 2 Imagine you have long but weakly overlapping image sequences (A overlaps with B, B with C, but A and C never see the same points, and so on. . .). Argue whether a pure factorization (using a single big measurement matrix) or a sequential approach is better suited here.

Solution: A pure factorization approach assumes strong visibility. If it is weak, we get large blocks of zeros, so the matrix is almost block diagonal, making the decomposition ill conditioned. So sequential approach is preffered here.